INCH-POUND
MIL-DTL-14970E
10 December 2018
SUPERSEDING
MIL-DTL-14970D
25 November 2003

DETAIL SPECIFICATION

COMPOSITION A-5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high explosive for use in ammunition.

1.2 Classification. Composition A-5 is of the following (see 6.1):

Class 1	-	Composition A-5 which has not been subjected to density/sensitivity test (see 4.5.7) may use not greater than 50 percent by weight of recovered/recycled RDX and not less than 50 percent by weight of virgin RDX in its feedstock for production (see 6.6).
Class 2	-	Composition A-5 which has passed the density/sensitivity test (see 4.5.7) consisting of only virgin RDX (see 6.1.1 and 6.6) in its feedstock for production.

Comments, suggestions, or questions on this document should be addressed to Commander, US Army ARDEC, RDAR-EIQ-SA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1376

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-271	-	Stearic Acid, Technical
MIL-DTL-398	-	RDX (Cyclotrimethylenetrinitramine)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-650	-	Explosive; Sampling, Inspection and Testing
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card

(Copies of these documents are available online at http://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL ORDNANCE SYSTEMS COMMAND DRAWINGS

LD549486	-	Explosive Properties, List of Drawings, Assemblies, Parts, Specifications, Etc
2426912	-	Explosive Properties Assembly
2426913	-	Donor Assembly
2426914	-	Acceptor Assembly
2426917	-	Attenuator

(Copies of these documents are available online at https://corp.dc3n.navy.mil)

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2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E203	-	Standard Test Method for Water Using Volumetric Karl Fischer Titration
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA, 19428-2959 or www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample of Composition A-5 shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample of Composition A-5 shall be subjected to conformance inspection in accordance with 4.3.

3.2 Material. Composition A-5 shall consist of RDX and shall be thoroughly and uniformly coated with stearic acid to form a homogeneous composition (see 6.5).

3.2.1 RDX. Composition A-5 shall consist of RDX, Type II, Class 1 (or Class 1 nominal) complying with the requirements of MIL-DTL-398.

3.2.1.1 Limitation on the use of recovered/recycled RDX in feedstock for Composition A-5. Composition A-5, Class 1 may be produced using RDX which has been recrystallized from a feedstock consisting of not greater than 50 percent by weight of recycled RDX from the RDX acid removal process only (see 6.6.2) and not less than 50 percent by weight of virgin RDX (see 6.6.1). Composition A-5, Class 2 may not be produced using recovered/recycled RDX.

3.2.2 Stearic acid. The stearic acid shall comply with the requirements of MIL-S-271.

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3.2.3 Composition. The composition shall be as specified in Table I, on a moisture free basis.

TABLE I. Composition.

	Composition A-5 weight percent Class 1 and/or Class 2
RDX, %	98.75 ± 0.25
Stearic Acid, %	1.25 ± 0.25

3.3 Moisture. The moisture content of the composition shall be not greater than 0.10 percent.

3.4 Solvent content. The solvent content of the composition shall be not greater than 0.30 percent cyclohexanone.

3.5 Insoluble particles. There shall not be insoluble particles retained on a No. 40 sieve and not greater than 5 insoluble particles retained on a No. 60 sieve when tested in accordance with 4.5.4.

3.6 Granulation. The composition shall comply with granulation requirements specified in Table II. All percentages shall be by weight using sieves conforming to ASTM E11. The powder shall pass through the required sieves readily without balling or the particles clinging together.

TABLE II. Granulation.

Through US Alternative Sieve No.	Composition A-5, Class 1 and/or Class 2	
	%, Min	%, Max
12	99.0	-
200	-	2.4

3.7 Bulk density. Composition A-5 shall have a bulk density not less than 0.95 g/cm^3.

3.8 Density/sensitivity (Class 2 only). Unless otherwise specified (see 6.1.1), Class 2 Composition A-5, shall have a density not less than 1.645 g/cm^3 and not greater than 1.730 g/cm^3 and an average not greater than 1.723 g/cm^3. When pressed into pellets the sensitivity of Class 2 Composition A-5 shall result in no explosion (dents shall be no greater than 0.0020 inches).

3.9 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.10 Energetic material qualification. Composition A-5 and all energetic materials or manufacturing processes used therein shall be qualified for their intended military use by the Army Qualification Authority or as specified in the contract (see 6.8).

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3.11 Workmanship. Composition A-5 shall be free of coarse gritty particles, visible impurities and foreign material such as dirt, rust, paint, metal chips or any other foreign objects so that the safety and reliability of the explosive are not compromised. The composition shall be thoroughly and uniformly coated with stearic acid.

4. VERIFICATIONS

TABLE III. Requirement/verification cross reference matrix.

Method of Verification 1- Analysis 2- Demonstration 3- Examination 4- Test		Classes of Verification A- First article B- Conformance						
Section 3 Requirement	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2.1	RDX			X		X	X	4.4.1
3.2.2	Stearic Acid			X		X	X	4.4.2
3.2.3	Composition	X		X	X	X	X	4.5.1
3.3	Moisture			X	X	X	X	4.5.2
3.4	Solvent content			X	X	X	X	4.5.3
3.5	Insoluble particles			X	X	X	X	4.5.4
3.6	Granulation			X	X	X	X	4.5.5
3.7	Bulk density	X		X	X	X	X	4.5.6
3.8	Density/sensitivity			X	X	X	X	4.5.7
3.9	Ammunition lot numbering			X		X	X	4.5.8
3.10	Energetic material qualification			X		X	X	4.5.9
3.11	Workmanship			X		X	X	4.5.10

4.1 Classification of inspection. The following types of inspections shall be conducted on this item.

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article. When specified, a sample of Composition A-5 shall be subjected to first article verification in accordance with 4.3.2. The testing shall apply for three (3) consecutive batches that have been produced by the manufacturer using the same production process,

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procedures, and equipment as will be used in regular production. All materials shall be obtained from the same sources as will be used in regular production.

4.2.1 First article rejection. If any item of the samples fail to comply with the first article requirements, the first article shall be rejected.

4.3 Conformance.

4.3.1 Lot formation. A lot shall consist of one or more batches from not greater than 30 consecutive calendar days of production of one composition and class designation only, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of composition and class that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain:

a. RDX of one type, from one interfix lot number and from one manufacturer only.

b. Stearic acid from one interfix lot number, from one manufacturer only.

4.3.2 Classification of characteristics.

4.3.2.1 Sampling. The tests described in 4.3.2.2 shall be performed on each batch on samples representative of the batch which were taken in accordance with ASTM E300, for solids. Approximately 750 grams of the composition shall be taken for each batch. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected.

4.3.2.2	Composition A-5	Requirement Paragraph	Inspection Method Reference
Critical	None defined		
Major			
101	RDX	3.2.1	4.4.1
102	Steric Acid	3.2.2	4.4.2
103	Composition	3.2.3	4.5.1
104	Moisture	3.3	4.5.2
105	Solvent content	3.4	4.5.3
106	Insoluble particle	3.5	4.5.4
107	Granulation	3.6	4.5.5
108	Bulk density	3.7	4.5.6
109	Density/sensitivity (Class 2 only)	3.8	4.5.7
Minor			
201	Ammunition lot numbering	3.9	4.5.8
202	Workmanship	3.11	4.5.10

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4.4 Materials.

4.4.1 RDX. Verify by objective evidence that RDX complies with MIL-DTL-398, Type II, Class 1 (or Class 1 nominal).

4.4.2 Stearic Acid. Verify by objective evidence that stearic acid complies with MIL-S-271.

4.5 Test methods and procedures. Unless otherwise specified herein, all chemicals and reagents shall be ACS or reagent grade. (See 6.3)

4.5.1 Determination of composition.

4.5.1.1 Percent stearic acid.

a. Place an approximately 5.00 g portion, weighed to four decimal places, of the dried sample in a tared, clean, dry, medium porosity filtering crucible containing a glass fiber filter pad. Place the crucible containing the sample into a stainless steel pan.

b. Add 20 mL of aliphatic naphtha (see 6.4) saturated with RDX into the crucible and allow the solvent to remain in contact with the explosive, at room temperature, for approximately 5 minutes. Stir the contents of the crucible to break up lumps and to aid in extraction of the sample. Perform all extractions and dispensing of naphtha under a fume hood.

c. Remove the crucible from the pan and place it on a vacuum source to filter off the solvent. Repeat the addition of 20 mL of naphtha, extraction for 5 minutes with occasional stirring and filtering, two additional times.

d. Wash the interior of the crucible with approximately 20 mL of naphtha from a wash bottle. Aspirate the residue until the naphtha solvent is no longer detectable (approx. 2 minutes).

e. Dry the crucible and residue in a drying oven or vacuum dryer at 90°C to 105°C for approximately 10 minutes.

f. Remove the crucible and residue from the oven and allow it to cool to room temperature in a desiccator and record the weight to four decimal places. Calculate the percent of stearic acid on a moisture free basis as follows:

$$\%\ Stearic\ Acid = \frac{100(X - Y)}{W}$$

Where:

X = initial weight of crucible and contents
Y = final weight of crucible and contents
W = weight of specimen on a moisture free basis

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4.5.1.2 Percent RDX.

a. Place the residue from the determination of stearic acid in a 200 mL beaker and add 100 ml of acetone.

b. Cover the beaker with a watch glass and heat the beaker and contents on a steam bath until the material is dissolved.

c. Filter the contents of the beaker through a tared medium filtering crucible. The crucible shall have been previously prepared by washing with acetone, igniting and weighing.

d. Care shall be taken to transfer all insoluble matter into the crucible.

e. Wash the residue 3 times with 30 ml portions of acetone. Dry the crucible for 30 minutes in an oven at 105ºC + 5ºC, cool in a desiccator and weigh.

f. Calculate the percent of RDX as follows:

$$\% \, RDX = \frac{100(Y - Z)}{W}$$

Where:

Y = Final weight of crucible and contents from the determinations of stearic acid
Z = Final weight of crucible and contents
W = Weight of original sample on a moisture-free basis

4.5.2 Determination of moisture content. The moisture content shall be determined in accordance with ASTM E203, with methanol as the special solvent. The sample size shall be based on the recommended sample amount from ASTM E203 which states one to five grams for water content in the 0.1% range.

4.5.2.1 Determination of moisture content (alternate). The moisture content shall be determined in accordance with Method 101.4 of MIL-STD-650, with methanol as the special solvent. Use a five gram sample for this determination.

4.5.3 Solvent content.

a. The cyclohexanone content is determined with a gas chromatograph (GC) equipped with a column that yields suitable peak shape, resolution, and retention (e.g. DB-1, 15m x 0.320mm) and detector suitable for cyclohexanone.

b. Prior to performing the cyclohexanone content determination, a calibration curve is prepared with known cyclohexanone concentrations. A standard stock solution is prepared by pipetting 1.0-mL of high-purity (HPLC-grade) cyclohexanone into a 250-mL volumetric flask and dilute to 250-mL with HPLC-grade acetone.

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c. The calibration curve is made by taking appropriate dilutions from the standard stock solution as follows:

Standard #1 (40-ppm of cyclohexanone)
Pipette 1.0-mL of the stock solution into a 100-mL volumetric flask and dilute to 100-mL with HPLC-grade acetone.

Standard #2 (80-ppm of cyclohexanone)
Pipette 2.0-mL of the stock solution into a 100-mL volumetric flask and dilute to 100-mL with HPLC-grade acetone.

Standard #3 (200-ppm of cyclohexanone)
Pipette 5.0-mL of the stock solution into a 100-mL volumetric flask and dilute to 100-mL with HPLC-grade acetone.

d. The resulting correlation factor for the calibration curve shall be 0.99 or greater to be considered a valid calibration curve.

e. Due to the volatile nature of the standards, fresh standards shall be made each day of use and stored in sealed containers.

f. The GC parameters (e.g. carrier gas pressure, injection temperature, oven temperature, and injection volume) may be adjusted to maintain adequate resolution and shape of the cyclohexanone peak.

g. An approximate 5.0 g sample of Composition A-5 is weighed, recorded to four decimal places, and transferred to a 100-mL volumetric flask. Approximately 80-mL of HPLC-grade acetone is added in order to dissolve the sample.

h. After the sample has dissolved, HPLC-grade acetone is added to the 100-mL mark and the flask is inverted several times to thoroughly mix the solution.

i. The sample is injected into the GC. The percent of cyclohexanone is calculated by the following equation:

$$\%\ Cyclohexanone = \frac{Cyclohexanone\ ppm}{Sample\ mass * 10{,}000} * 100$$

4.5.4 Insoluble particles. The insoluble particles shall be determined in accordance with MIL-STD-650, Method 106.1 with acetone as the solvent.

4.5.5 Granulation. Determine the granulation in accordance with Method 204.1 of MIL-STD-650 except with a shaker geared to produce 300 +15/-32 gyrations and sieves shall be in accordance with ASTM E11. Failure to meet the granulation shall result in batch rejection.

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4.5.6 Bulk density. The bulk density shall be determined using a 200 gram sample with a 250 mL graduated cylinder and a 4 inch by 4 inch by 4 inch plywood box which is padded on the inside surfaces with ¼ inch rubber foam having a density of approximately 0.029g/mL. The top of the box shall have a circular hole cut in the center of sufficient diameter so that the cylindrical portion of the graduate fits easily through it. To determine the bulk density of the sample pass the cylindrical portion of the graduated cylinder through the hole in the top of the box so that the base of the graduate is in contact with the inside of the box top. Place the top on the box enclosing the bottom portion of the graduated cylinder inside the box. Place the box on a firm flat surface and transfer an accurately weighed to the nearest 0.1g portion of the sample to the graduate. Lift the protruding portion of the graduate until the base comes into contact with the top of the box and release allowing the base of the graduate to strike the padded bottom of the box. Repeat the lifting and dropping procedure until twenty drops have been completed. Read the volume of the compacted powder in the cylinder to the nearest mL. Calculate bulk density in grams per mL, as follows:

$$Bulk\ Density = \frac{A}{B}$$

Where:

A = weight of powder (g)
B = volume of compacted powder (mL)

4.5.7 Density/sensitivity test (Class 2 only). (See Dwg. LD549486) A representative sample of Explosive Composition A-5 from 4.3.2.1 shall be subjected to the procedures of the following paragraphs:

4.5.7.1 Loading and calibration of donor assemblies. A minimum batch of twenty-five donor assemblies shall be prepared in accordance with NAVORD Drawing 2426913. These donors shall be assembled in the test arrangement and fired against the block by initiation of the detonator with a 50 volts DC (minimum) 20 amperes (minimum) power supply. To be acceptable for use in the sensitivity test, the average depth of dent produced in the block by the donors must be between 0.063 and 0.069 inch and the standard deviation must not exceed 0.005 inch. Each block shall be used once and only once. The measurement of the indentation depth shall be made in accordance with 4.5.7.5.

4.5.7.2 Preparation of acceptor specimens and measurement of density. The Explosive Composition A-5 shall be loaded in eight equal weight increments and each consolidated in accordance with Drawing 2426914. The first trial loading shall be with increment weights of 0.16 g of the explosive. The acceptor body (NAVORD Drawing 2426914) shall be weighed before and after loading. If all eight increments fit in the acceptor body with room to spare, adjust the subsequent acceptor by measuring the remaining unloaded column height and increasing equally the weight of each increment to meet the tolerance shown in NAVORD Drawing 2426914. If all eight increments do not fit into the acceptor body, adjust equally the individual increment weights based on the actual weight of explosive contained in the body such that the tolerances shown in NAVORD Drawing 2426914 are obtained. Load another test body to assure that the drawing tolerance has been met. When the adjustment is satisfactory, load the

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acceptor bodies to form a total of exactly 20 acceptors meeting the tolerance shown in NAVORD Drawing 2426914. The acceptors shall be weighed before and after loading and the density of each individual acceptor shall be determined and reported to three decimal places. The individual densities shall be between 1.645 g/cm^3 and 1.730 g/cm^3 and the average density shall not exceed 1.723 g/cm^3.

4.5.7.3 Sensitivity test assemblies. Twenty explosive properties assemblies shall be prepared in accordance with NAVORD Drawing 2426912 from a random selection of the acceptable donors prepared in accordance with paragraph 4.5.7.1and the acceptors prepared in accordance with paragraph 4.5.7.2. The concentricity of the acceptor to the dent block shall be within 0.250 inch. The longitudinal axis of the donor, acceptor and barrier shall be made concentric by taping with two wraps of masking tape around the periphery of these pieces to form a single unit.

4.5.7.4 Test procedure. Twenty and only twenty assemblies shall be tested using 3.5 decibang attenuators (NAVORD Drawing 2426917 Pc. No. 41). The dents produced in the witness blocks shall be measured in accordance with paragraph 4.5.7.5.

4.5.7.5 Measurement of indentation depth. Depth of indentation made in the block by the explosion of the donor or acceptor as applicable, shall be measured with a indicator accurate to 0.0001 inch or better. The point of the dial indicator probe shall have an approximate 30 degree included angle and the end of the point shall have a radius of 0.025 + 0.002 inch. Before measuring the depth of indentation in the block, remove any foreign material, such as deposits, from the dent. Zero the indicator with the point of the probe in the deepest part of the dent. Then take the readings on the surface of the block at four points near the periphery of the block. These points shall be approximately 0.125 inch away from the periphery and 90 degrees apart. The dent shall be computed as the average of the four readings.

4.5.7.6 Acceptance criteria. The explosive shall be considered to have passed the sensitivity test, if there are no acceptor explosions in the 20 trials. Any reaction causing a dent of 0.0020 inch or more shall be rejected.

4.5.8 Ammunition lot numbering. Visually verify that an ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.5.9 Energetic material qualification. Verify by examination of objective evidence that the Army Qualification Authority has qualified all energetic materials for use (see 6.8).

4.5.10 Workmanship. Take approximately 200 to 300 grams of the material and spread the composition out on a clean sheet of white paper and examine visually for the presence of visible impurities, extraneous foreign matter, and workmanship. Failure to meet the workmanship inspection and/or visual inspection of workmanship during the test methods of 4.5 shall result in batch rejection.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Composition A-5 is intended for use as a main charge, a lead, or booster explosive.

6.1.1 Navy acquisition. If specified in the contract or purchase order Composition A-5 for Navy acquisition may be supplied from those batches which have been subjected to and have passed the requirement of the density/sensitivity test (see 4.5.8). The Naval Sea Systems Command, Naval Ordnance Safety and Security Activity (NOSSA N85), restricts Navy procurement to class 2 only.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Requirements for submission of first article sample.
c. Class required (see 6.1).
d. Special marking, when specified.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product (if applicable).
g. Acceptance Inspection Equipment (AIE) requirements.
h. Acceptance and description sheet for propellants and explosives in accordance with MIL-STD-1171.
i. Certificate of Analysis.
j. Packaging requirements (see 6.9).
k. Ammo Data Cards IAW MIL-STD-1168.

6.2 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the technical agency (see 6.7). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

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6.3 Aliphatic naphtha. Aliphatic naphtha used in testing should have a boiling point above 100°C.

6.4 Process. The following wet slurry method has been found satisfactory. Molten stearic acid is added concurrently with cyclohexanone to a RDX water slurry in which the RDX was previously recrystallized with cyclohexanone. The solvent is then removed and the Composition A-5 slurry is cooled, filtered and dried.

6.5 Virgin RDX and recycled RDX.

6.5.1 Virgin RDX. Feedstock (crude RDX) to recrystallizer is composed of slurried RDX coming directly from the RDX synthesis process.

6.5.2 Recovered/recycled RDX. Feedstock (crude RDX) to recrystallizer is composed of slurried RDX recovered from (1) the HMX or RDX acid removal process and (2) the acetic acid recovery process.

6.6 Technical agency. The technical agency responsible for the preparation of this specification is ARDEC, ATTN: RDAR-EIQ-MG, Picatinny, NJ 07806 and ARDEC, ATTN: RDAR-MEE-W, Rock Island, IL 61299.

6.7 Energetic material qualification. When a contract is awarded to a manufacturer of Composition A-5 for the Government, qualification of energetic materials by the Army Qualification requires demonstration that the energetic material meets the requirements of AOP-7 and STANAG 4170. Point of contact for the Army Qualification Authority and list of qualified sources can be obtained from the Technical Agency (see 6.6).

6.8 Packaging requirements. Drawing 7548645 may be used to package Composition A-5.

6.9 Subject term (key word) listing.

RDX
Cyclotrimethylenetrinitramine
Stearic acid

6.10 Incorporated changes. This revision of the specification incorporates ECP R18Q2017.

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army - AR
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Preparing Activity:
Army- AR
(Project 1376-2019-002)

Review activities:
Army-AV, MI, TE
Navy- AS
Air Force-11, 70

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